

5.1.3 Acids, bases and buffers

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Brønsted–Lowry acids and bases	
(a) (i) a Brønsted–Lowry acid as a species that donates a proton and a Brønsted–Lowry base as a species that accepts a proton (see also 2.1.4 Acids) (ii) use of the term <i>conjugate acid–base pairs</i> (iii) monobasic, dibasic and tribasic acids (b) the role of H^+ in the reactions of acids with metals and bases (including carbonates, metal oxides and alkalis), using ionic equations (see also 2.1.4 c, 2.1.5 e) (c) (i) the acid dissociation constant, K_a, for the extent of acid dissociation (see also 2.1.4 b) (ii) the relationship between K_a and pK_a	Learners should be able to identify acid–base pairs in equations for acid–base equilibria. HSW1,7 Development of different models over time to explain acid–base behaviour.
pH and $[H^+(aq)]$	
(d) use of the expression for pH as: $pH = -\log[H^+]$ $[H^+] = 10^{-pH}$	M0.1, M0.4, M2.2, M2.3, M2.4, M2.5 HSW8 pH as convenient terminology for communicating $[H^+]$.
(e) use of the expression for the ionic product of water, K_w	M0.1, M0.4, M2.2, M2.3, M2.4
(f) calculations of pH, or related quantities, for: (i) strong monobasic acids (ii) strong bases, using K_w	M0.1, M0.4, M2.2, M2.3, M2.4, M2.5
(g) calculations of pH, K_a or related quantities, for a weak monobasic acid using approximations	M0.1, M0.4, M2.1, M2.2, M2.3, M2.4, M2.5 Approximations for weak acid calculations: $[HA]_{\text{equilibrium}} \sim [HA]_{\text{undissociated}}$ i.e. $[HA] \gg [H^+]$ $[H^+]_{\text{equilibrium}} \sim [A^-]_{\text{equilibrium}}$ i.e. negligible dissociation of H_2O. Learners will not be required to solve quadratic equations.

- (h) limitations of using approximations to K_a related calculations for 'stronger' weak acids

M0.3

Including reasons why

$[HA]_{\text{equilibrium}} \sim [HA]_{\text{undissociated}}$ may no longer be valid.

HSW6 Understanding of the circumstances under which K_a approximations break down.

Buffers: action, uses and calculations

- (i) a buffer solution as a system that minimises pH changes on addition of small amounts of an acid or a base
- (j) formation of a buffer solution from:
- (i) a weak acid and a salt of the weak acid, e.g. $\text{CH}_3\text{COOH}/\text{CH}_3\text{COONa}$
 - (ii) excess of a weak acid and a strong alkali, e.g. excess $\text{CH}_3\text{COOH}/\text{NaOH}$
- (k) explanation of the role of the conjugate acid–base pair in an acid buffer solution, e.g. $\text{CH}_3\text{COOH}/\text{CH}_3\text{COO}^-$, in the control of pH
- (l) calculation of the pH of a buffer solution, from the K_a value of a weak acid and the equilibrium concentrations of the conjugate acid–base pair; calculations of related quantities
- (m) explanation of the control of blood pH by the carbonic acid–hydrogencarbonate buffer system

M0.1, M0.4, M2.2, M2.3, M2.4, M2.5

The $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ buffer is present in blood plasma, maintaining a pH between 7.35 and 7.45.

Neutralisation

- (n) pH titration curves for combinations of strong and weak acids with strong and weak bases, including:
- (i) sketch and interpretation of their shapes
 - (ii) explanation of the choice of suitable indicators, given the pH range of the indicator
 - (iii) explanation of indicator colour changes in terms of equilibrium shift between the HA and A^- forms of the indicator
- (o) the techniques and procedures used when measuring pH with a pH meter.

M3.1

No indicator is suitable for a weak acid/weak base titration.

The indicator should be considered as a weak acid, HA.

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HSW4 Opportunities to carry out experimental and investigative work.